

Tutorial 1 – Solutions

Q1. Swapping values of 2 variables.

Method 1

```
## Swapping Two values
## Tutorial questions, IP

## Method 1
## - using extra variable

print("Before Swapping-->")

value1 = int(input("Enter value1: "))
value2 = int(input("Enter value2: "))

temp = value2 ## value2 copied to a third variable
| | | ## called 'temp'

value2 = value1 ## the data stored in value1
| | | ## copied to value2

value1 = temp ## 'temp' which consisted value2 data,
| | | ## it's content has been copied to value1

print("After Swapping --> value1:", value1)
print("After Swapping --> value2:", value2)
```

Output 1

```
Before Swapping-->
Enter value1: 5
Enter value2: 7
After Swapping --> value1: 7
After Swapping --> value2: 5
```

Method 2

```
## Swapping Two values
## Tutorial questions, IP

## Method 2
## - without using extra variable

print("Before Swapping-->")

value1 = int(input("Enter value1: "))
value2 = int(input("Enter value2: "))

value1 = value1 + value2
value2 = value1 - value2
value1 = value1 - value2

print("After Swapping --> value1:", value1)
print("After Swapping --> value2:", value2)
```

Output 2:

```
Before Swapping-->
Enter value1: 11
Enter value2: 41
After Swapping --> value1: 41
After Swapping --> value2: 11
```

Q2. Take input, a number (X) from the user. Take its multiple (X*Y) [X and Y can be arbitrary numbers]. Return the product.

Does it work?

```
## TYPE-Casting
## Tutorial 1, IP

X = input("Enter number: ")
Y = 4
wrong_product = Y * X

print("Result of multiplying without typecasting:")
print("Wrong Product= ", wrong_product )
## This product is concatenation of string X, Y many times.

print("Let's Take the input, the correct way: ")

X = int(input("Enter number again: "))

correct_product = Y * X

print("Correct Product= ", correct_product )
```

Output

```
Enter number: 5
Result of multiplying without typecasting:
Wrong Product=  5555
Let's Take the input, the correct way:
Enter number again: 5
Correct Product=  20
```

Q3. Use of formatted strings.

Suppose we have to print the value of two defined variables using the print() function.

V1 = 12 // clearly an int (inputted)

V2 = 30.0 // a real value (inputted)

```
## Use of formatted string
## Tutorial 1, IP

## Let's take two inputs V1(an integer) and V2(a float/real value)

V1 = int(input("Enter V1: "))
V2 = float(input("Enter V2: "))

## Let's print these values using simpler syntax

print("Value stored in V1 is:", V1, "and Value stored in V2 is:", V2)

## Now we can print the same thing using:->

print("Value stored in V1 is: {} and Value stored in V2 is: {}".format(V1, V2))

## same thing can also be printed using :->

print( f"Value stored in V1 is: {V1} and Value stored in V2 is: {V2}" )
## Notice the prefix 'f' here
```

Output

```
Enter V1: 12
Enter V2: 30.5
Value stored in V1 is: 12 and Value stored in V2 is: 30.5
Value stored in V1 is: 12 and Value stored in V2 is: 30.5
Value stored in V1 is: 12 and Value stored in V2 is: 30.5
```